Cars 2 Has an Alibi

Team Members: _____ _____ _____

Twenty-one Pixar films, each rated by three different services. Today you will accuse a movie of ruining the data. The evidence against it is real — but the test you run will not convict it, and working out why is the point.

Open `Class17_Cars_2_Has_An_Alibi_Skeleton.ipynb`. Answer here, on paper.

Part 1. One Variable at a Time

18 minutes

Before comparing two variables you have to understand each one alone. The `stats()` method gives you six numbers per column.

> **1.1 Fill in from the notebook:**

	rotten tomatoes	**metacritic**	**critics choice**
mean			
median			
mean − median			

> **1.2 One of those three columns has a mean far below its median; the other two barely differ. Which one, and what does a mean well below the median tell you about the shape of a distribution? (Sketch the shape if it helps.)**

Now some vocabulary you need before the next plot. Order the values low to high and cut them into quarters. *Q1* is the median of the lower half, *Q2* is the median, *Q3* is the median of the upper half. The distance from Q1 to Q3 is the *interquartile range*, or IQR.

> **1.3 *By hand, no code.* For these twelve numbers:**
>
> **24, 25, 26, 27, 30, 32, 40, 44, 50, 52, 55, 57**
>
> **Q1 =** _____ **Q2 =** _____ **Q3 =** _____ **IQR =** _____

A *box plot* draws exactly that: a box from Q1 to Q3 with a line at the median. The whiskers reach the farthest point within 1.5 × IQR of the box, and anything past a whisker is drawn as a separate dot — a *flier*.

> **1.4 Run `rmc.boxplot()`.**
>
> **Which service gives the most favorable ratings overall?** _____

The least favorable? _____

How many fliers appear in the whole plot? _____ **On which service?** _____

1.5 Find that film in the table and write down its name and its three ratings. Judging only by the box plot, would you call it an outlier? Say why.

Part 2. Two Variables at a Time

18 minutes

2.1 Look at the scatter plot of `metacritic` against `critics_choice`. Is there a correlation? Positive or negative? Strong, moderate, or weak? Commit to an answer before you compute anything.

Standardizing turns any measurement into standard units — how many SDs from its own mean:

$$z = \frac{x - \bar{x}}{\sigma}$$

2.2 Plot the standardized ratings against each other. Compare the shape of the cloud to the unstandardized version. What changed, and what did not? Why?

The correlation coefficient r is the *average of the product* of the two variables in standard units. It has no units and always falls between -1 and $+1$.

2.3 Let x = `metacritic` and y = `critics_choice`. For one film, think about the product $z_x \times z_y$. This is about *signs only* — you need no numbers from the notebook. Write + or − in each cell:

	y above its mean ($z_y > 0$)	**y below its mean ($z_y < 0$)**
x above its mean ($z_x > 0$)		
x below its mean ($z_x < 0$)		

Each cell is one quadrant of your standardized scatter plot from 2.2, split by the lines at 0. Which quadrants hold most of the 21 films? Then explain in one sentence why r comes out large and positive when two variables track each other, and near zero when they do not.

2.4 Compute r for all three pairs and rank them.

`rotten_tomatoes` **&** **metacritic** $r =$ _____

`rotten_tomatoes` **&** `critics_choice` $r =$ _____

metacritic & `critics_choice` $r =$ _____

Which two services agree with each other most closely? Does that match what you guessed in 2.1?

Part 3. The Alibi

14 minutes

The film from Part 1 sticks out on the box plot and sticks out on the scatter plot. It is the obvious candidate for the point distorting your correlation. Put it on trial.

3.1 For `rotten_tomatoes` vs. `metacritic`:

r with all 21 films _____ **r with Cars 2 removed** _____

Before computing, predict: will dropping Cars 2 strengthen or weaken r, and by a lot or a little? Then check. Which part of your prediction held up?

3.2 The notebook removes each film one at a time and reports how much r changes.

Largest effect: _____ **change** _____ **Cars 2′s change** _____

Find that film on the scatter plot and mark it. Describe where it sits compared to Cars 2.

Cars 2 scored 57 on metacritic. The other low-rated films suggest a score near **46** would have been in keeping with the pattern. The notebook tries that hypothetical.

3.3 Record all three:

r **as observed** _____ r **if Cars 2 had scored 46** _____ r **with Cars 2 removed** _____

Being out of line was worth about 0.06 **of correlation. But removing Cars 2 gained only about** 0.02**. Roughly** 0.04 **went somewhere. Compare the spread of** `rotten_tomatoes` **with and without Cars 2, and explain where it went.**

3.4 So: is Cars 2 an unusual point? Is it the point that most affects r**? Explain how both answers can be what they are, in one or two sentences.**

Four small data sets, all with the same means, the same standard deviations, and the same correlation, $r \approx 0.816$ — close to the correlation you found in 2.4.

3.5 Plot all four. Which of them would you be comfortable summarizing with a correlation coefficient alone, and which would be actively misleading? Give a reason for each one you rule out.